AVERAGE-BASED WATERMARK FORGERY

Name: Rohit Chandrakant Ahire
Matriculation Number: 7069699

Name: Abdul Habeeb
Matriculation Number: 7070246

Team-ID:- `team_LXXVI`
GitHub Repository:- `https://github.com/AhireRohit/TML_Assignment_4`

1 INTRODUCTION

In this assignment, the goal was to forge invisible watermarks on clean target images. We were given 200 clean target images and 8 groups of watermarked source images. Each group, from `WM_1` to `WM_8`, contained 25 images watermarked with the same hidden message, but the actual watermarking method was unknown. The task was to slightly change the clean target images so that the detector would detect the correct watermark, while the images should still look almost the same as the original clean images. Since the final score depends on both watermark detection and LPIPS image quality, the attack needs to be strong enough but not too visible.

2 METHOD

Our final method is based on a simple average-copy idea. The main assumption is that all images inside one watermark folder share the same hidden watermark signal, while the visible image content is different. Because of this, averaging the watermarked source images can reduce image-specific content and keep some of the common watermark signal.

For each watermark group `WM_i`, we computed the average watermarked source image:

$$A_i = \frac{1}{25}\sum_{j=1}^{25} W_{i,j},$$

where $W_{i,j}$ is the j-th watermarked source image from group i. For a clean target image x, we used the average source image to create the main perturbation:

$$\Delta_{\text{avg}} = \lambda_i(A_i - x).$$

This means that the target image is moved slightly towards the average watermarked image of the correct group.

The value of λ was important because the score depends on both watermark detection and visual quality. A small λ keeps LPIPS low, but the watermark signal becomes weak. A large λ gives a stronger watermark signal, but it also changes the image too much and hurts the LPIPS part of the score. In our experiments, $\lambda = 0.16$ gave the best global balance. Lower values were weaker, while stronger values did not improve the public score further.

We also used a small high-frequency residual component. For this, each watermarked source image was denoised using non-local means denoising. Then we subtracted the denoised image from the original watermarked image:

$$R_i = \text{trimmed\_mean}_j(W_{i,j} - D(W_{i,j})),$$

where $D(\cdot)$ is the denoising operation. This residual helps because invisible watermarks are often hidden as small high-frequency changes. However, using only this residual was not enough, so we used it as a support signal together with the average-image perturbation.

The final forged image was generated as:

$$\hat{x} = \text{clip}(x + \Delta_{\text{avg}} + \beta_i R_i).$$

Clipping was also important. Without clipping, some images changed too much in smooth regions. With clipping, the perturbation stays limited, which helped to keep the images visually closer to the clean targets. The best global setting used $\lambda = 0.16$, residual strength $\beta = 2.0$, low clipping threshold 30, total clipping threshold 40, and texture mask strength 0.12.

After getting the best global result, we performed a small per-watermark ablation. We found that `WM_1` worked better with a stronger average-mix setting. For `WM_1`, we used $\lambda = 0.20$, residual strength $\beta = 1.0$, low clipping threshold 32, and total clipping threshold 42. The final submission therefore uses the global setting for most watermark groups and uses the stronger setting only for the `WM_1` batch.

3 Results

Table 1 shows the main public leaderboard results from our experiments. The residual-only method gave a useful starting point, but the average-mix method was clearly better.

Table 1: Main public leaderboard results.

Method	**Public Score**
NLM residual only	0.387863
Average-mix, $\lambda = 0.08, \beta = 4.0$	0.412935
Average-mix, $\lambda = 0.12, \beta = 3.0$	0.436932
Average-mix, $\lambda = 0.16, \beta = 2.0$	0.442007
Final with `WM_1` ablation	**0.444834**

The residual-only method reached 0.387863. Increasing the residual strength did not improve the score, which suggests that the added high-frequency signal was either not strong enough for the detector or was hurting visual quality.

The average-mix method gave the main improvement. With $\lambda = 0.08$, the score improved to 0.412935. Increasing λ to 0.12 improved the score to 0.436932, and $\lambda = 0.16$ gave the best global score of 0.442007. This shows that the average source image carried a transferable watermark signal. However, when we increased the strength further, the public score did not improve. This is likely because the detector gain was cancelled by the LPIPS penalty.

Finally, we tried a per-watermark replacement. We replaced only the `WM_1` batch with a stronger average-mix variant using $\lambda = 0.20$ and $\beta = 1.0$. This gave our best public leaderboard score:

$$\mathbf{0.444834}.$$

Replacing `WM_2` with the same stronger variant did not improve the score further, so the final submission keeps only the `WM_1` replacement.

4 Conclusion

This assignment shows that invisible watermarking can be weak against copy attacks when an attacker has several examples with the same hidden message. Even without knowing the watermarking algorithm, averaging watermarked examples can reveal a transferable signal. By adding a small part of this signal to clean target images, it was possible to cause false watermark detection while keeping the images visually close to the originals.

This matters in real systems because watermark detectors may be used for ownership claims, image provenance, or checking whether an image came from a specific model. If an attacker can forge such a watermark, then an unrelated image could be falsely linked to a model or owner. This can make watermark-based evidence less reliable. So watermarking systems should be tested against copy attacks, avoid reusing the same message too much, and should be combined with other ways of proving image origin.